

Module 2 Lesson 5

Linear Inequalities in One Variable (6.4)

Objectives

In this lesson, we will learn how to

- graph subsets of real numbers on a number line;
- solve linear inequalities;
- solve applied problems using linear inequalities.

1. Linear Inequalities

A **linear inequality** in one variable is similar to a linear equation, but it uses an inequality symbol instead of an equal sign.

A linear inequality may look like

$$ax + b \leq c,$$

where the inequality symbol can be

$$<, >, \leq, \geq.$$

Linear Inequality

A linear inequality in one variable can be written using one of the forms

$$ax + b < c,$$

$$ax + b > c,$$

$$ax + b \leq c,$$

or

$$ax + b \geq c.$$

Solving an inequality means finding all numbers that make the inequality true.

The set of all numbers that satisfy an inequality is called the **solution set**.

2. Inequality Symbols

Common Inequality Symbols

Symbol	Meaning	Number Line Point
$<$	less than	open circle
$>$	greater than	open circle
$\leq$	less than or equal to	closed circle
$\geq$	greater than or equal to	closed circle

Open and Closed Circles

- Use an **open circle** for $<$ or $>$.
- Use a **closed circle** for $\leq$ or $\geq$.

3. Graphing Subsets of Real Numbers

We can graph solution sets on a number line.

For example,

$$\{x \mid x < 3\}$$

means all real numbers less than 3.

$$\{x \mid x \geq -1\}$$

means all real numbers greater than or equal to -1 .

$$\{x \mid -1 < x \leq 3\}$$

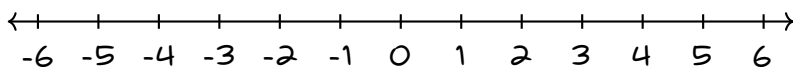
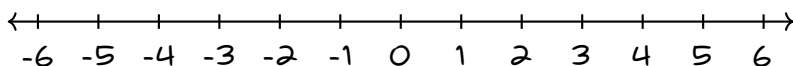
means all real numbers greater than -1 and less than or equal to 3.

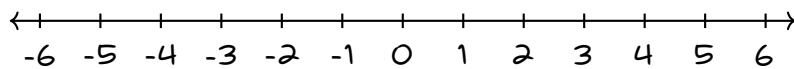
Example 1: Graph each set on a number line:

$$\{x \mid x < 3\}$$

$$\{x \mid x \geq -1\}$$

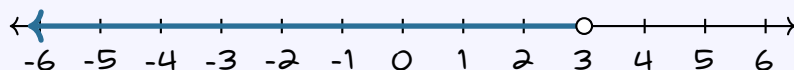
$$\{x \mid -1 < x \leq 3\}$$



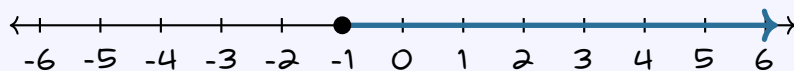


Solution

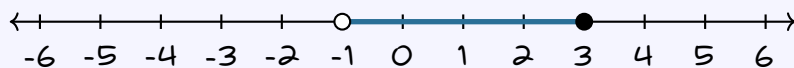
For $\{x \mid x < 3\}$, use an open circle at 3 and shade left.



For $\{x \mid x \geq -1\}$, use a closed circle at -1 and shade right.



For $\{x \mid -1 < x \leq 3\}$, use an open circle at -1 , a closed circle at 3, and shade between.



4. Solving Linear Inequalities

The procedure for solving linear inequalities is almost the same as the procedure for solving linear equations.

There is one very important exception.

Important Rule

When multiplying or dividing both sides of an inequality by a negative number, reverse the direction of the inequality symbol.

For example,

$$-2x < 8.$$

Divide both sides by -2 . Since we divide by a negative number, reverse the inequality symbol:

$$x > -4.$$

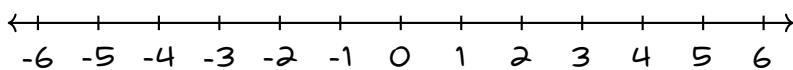
Solving a Linear Inequality

1. Simplify each side.
2. Collect variable terms on one side.
3. Collect constants on the other side.
4. Isolate the variable.
5. If multiplying or dividing by a negative number, reverse the inequality symbol.
6. Graph the solution set if requested.

5. Solving and Graphing a Linear Inequality

Example 2: Solve and graph the solution set:

$$6x - 12 > 8x + 2.$$



Solution

Start with

$$6x - 12 > 8x + 2.$$

Subtract $8x$ from both sides:

$$-2x - 12 > 2.$$

Add 12 to both sides:

$$-2x > 14.$$

Divide both sides by -2 . Reverse the inequality symbol:

$$x < -7.$$

The solution set is

$$\{x \mid x < -7\}.$$

Graph: open circle at -7 , shade left.

6. Three-Part Inequalities

A three-part inequality has two inequality symbols.

For example,

$$-3 < 2x + 1 \leq 3.$$

This means

$$2x + 1$$

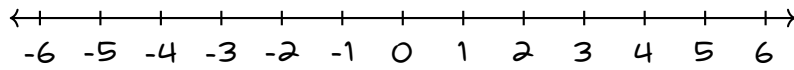
is greater than -3 and less than or equal to 3 .

Solving a Three-Part Inequality

To solve a three-part inequality, perform the same operation on all three parts.

Example 3: Solve and graph the solution set:

$$-3 < 2x + 1 \leq 3.$$



Solution

Start with

$$-3 < 2x + 1 \leq 3.$$

Subtract 1 from all three parts:

$$-4 < 2x \leq 2.$$

Divide all three parts by 2:

$$-2 < x \leq 1.$$

The solution set is

$$\{x \mid -2 < x \leq 1\}.$$

Graph: open circle at -2 , closed circle at 1 , and shade between.

7. Application of Linear Inequalities

Linear inequalities are useful when a problem involves words such as:

at least, at most, more than, less than.

Common Translations

Phrase	Symbol
less than	$<$
greater than / more than	$>$
at most / no more than	$\leq$
at least / no less than	$\geq$

8. Applied Problem

Example 4: A student has scores of 72, 85, and 78 on the first three tests.

What score must the student earn on the fourth test to have an average of at least 80?

Solution

Let

$x =$ the fourth test score.

The average must be at least 80:

$$\frac{72 + 85 + 78 + x}{4} \geq 80.$$

Simplify:

$$\frac{235 + x}{4} \geq 80.$$

Multiply both sides by 4:

$$235 + x \geq 320.$$

Subtract 235:

$$x \geq 85.$$

The student must earn at least

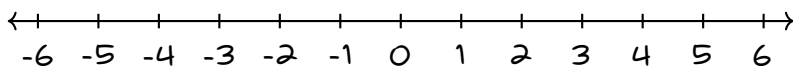
85

on the fourth test.

9. More Practice

Example 5: Solve and graph the solution set:

$$4x + 7 \leq 2x - 9.$$



Solution

Subtract $2x$ from both sides:

$$2x + 7 \leq -9.$$

Subtract 7:

$$2x \leq -16.$$

Divide by 2:

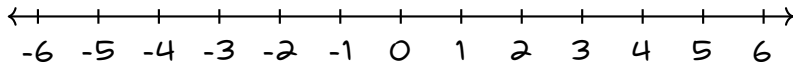
$$x \leq -8.$$

The solution set is

$$\{x \mid x \leq -8\}.$$

Example 6: Solve and graph the solution set:

$$-5x + 3 < 18.$$



Solution

Subtract 3 from both sides:

$$-5x < 15.$$

Divide by -5 . Reverse the inequality symbol:

$$x > -3.$$

The solution set is

$$\{x \mid x > -3\}.$$

Summary

Key Ideas

- A linear inequality uses one of the symbols $<$, $>$, $\leq$, or $\geq$.
- The solution set is the set of all numbers that make the inequality true.
- Use an open circle for $<$ or $>$.
- Use a closed circle for $\leq$ or $\geq$.
- To solve a linear inequality, use steps similar to solving a linear equation.
- When multiplying or dividing both sides by a negative number, reverse the inequality symbol.
- For three-part inequalities, perform the same operation on all three parts.
- Linear inequalities can model real-world situations involving limits or requirements.